Dang Quang Thien Tran

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Research Interests

I study trustworthy machine learning. A model's answer deserves trust only when we can check it against the evidence behind it. My research asks when such a check is possible, and what a model should do when it is not. I have pursued these questions in vision-language models, information retrieval, causal discovery, and medical AI, through interpretability, benchmark design, and closed-form analysis.

Education

2025–2026 **M.S. in Computer Science**, University of Massachusetts Amherst, GPA 3.84/4.00

2021–2025 **B.S. in Computer Science, Minor in Mathematics**, University of Massachusetts Amherst
GPA 3.94/4.00, Cum Laude (top 25%), Dean's List in all 8 semesters

Publications

[1] EACL 2027 *under review* **Dang Quang Thien Tran**[*], Quang V. Dang[*], Vinamra Tyagi[*], Sai Soorya Rao Veeravalli[*], Trang Nguyen, Ryan A. Rossi, Franck Dernoncourt, Nedim Lipka, Koustava Goswami, Samyadeep Basu. *MultAttnAttrib: Training-Free Multimodal Attribution in Long Document Question Answering.* arXiv:2607.01420. ([*]equal contribution)

Research Experience

2026–present **Abstention in vision-language models when image and text conflict** (in preparation)
UMass Amherst

2026–present **Robustness of findings in continuous causal discovery** (NeurIPS 2027 workshop, in preparation)
Knowledge Discovery Lab, UMass Amherst; advised by Prof. David Jensen

2026–present **Share of undirected edges in sparse random DAGs** (UAI 2027 workshop, in preparation)
Independent research

2026 **MultAttnAttrib: training-free multimodal attribution** (EACL 2027, under review) [1]
Adobe Research; mentored by Samyadeep Basu

- Designed MultAttnAttrib, attributing VLM answers to source text and images via retrieval heads
- Found retrieval heads sparse and largely shared, with image heads concentrated in deeper layers
- Outperformed same-model prompting and RAG baselines, and GPT-5.4 on image attribution
- Introduced MultAttrEval, the first benchmark for multimodal attribution in long documents

Keywords: mechanistic interpretability, attributed generation, long-document QA, causal mediation analysis

2025 **Tool-augmented vision-language models for medical VQA**
BioNLP Lab, UMass Amherst; advised by Prof. Hong Yu, mentored by Hieu Tran

- Curated a 500+ question medical VQA benchmark from five datasets; baselined six VLMs
- Showed that a zero-shot ReAct agent calling medical vision tools adds 6 points over the VLM

Keywords: multimodal agents, open-vocabulary detection, image segmentation

2023 **Tight security of RSA Full-Domain-Hash** (ERSP National Conference, poster)
Early Research Scholars Program, UMass Amherst; advised by Prof. Adam O'Neill

- Conjectured a tight security reduction for RSA-FDH in the Algebraic Group Model

Keywords: provable security, digital signatures

Industry Experience

Summer 2025 — **CB/I Digital**, Remote
Data Scientist Intern

- Discovered that sales-pattern clustering merged occasional sellers with never-sold products
- Lowered new-product wMAPE 12% by re-ranking embedding matches on shape and magnitude

Keywords: intermittent and cold-start demand forecasting, anomaly detection, STL decomposition

Summer 2023 — **FPT Information System**, Hanoi, Vietnam
Machine Learning Engineer Intern

- Deployed two-stage extractive QA with hybrid retrieval and a fine-tuned cross-encoder reranker
- Reached 85% recall and 81% precision, with retrieval about 70% faster than the prior system

Keywords: information retrieval, dense retrieval, BM25, learning to rank

Summer 2022 — **MobiFone**, Can Tho, Vietnam
Computer Vision Engineer Intern

- Accelerated field extraction from scanned legal documents with non-standard templates by 20%

Keywords: document layout analysis, OCR, U-Net, fuzzy string matching

Projects

2026–present — **Ctrl F-ilm, zero-shot video moment search**

- Developed search that returns the video and timestamps for a text query, tested on 4,600 videos

Keywords: cross-modal retrieval, video corpus moment retrieval, temporal grounding

2025 — **Brain tumor segmentation**

- Raised BraTS2020 Dice over UNet++ (0.79 to 0.85) via residual blocks, ASPP and Dice + CE loss

Keywords: medical image segmentation, MRI, multi-scale features, class imbalance

Teaching

Fall 2023 — Undergraduate Course Assistant, Introduction to C Programming, UMass Amherst

Honors and Awards

2025–2026 — **Bay State Scholar**, UMass Amherst
$17,000 per year, about 40% of MS tuition

2021–2025 — **Chancellor's Award**, UMass Amherst
$16,000 per year, for top out-of-state admits

2021 — **Best Remote Health Hack**, HackHolyoke

Skills

Programming	Python, C/C++, Java, R, SQL, LaTeX
ML, science	PyTorch, vLLM, nnsight, scikit-learn, NumPy, SciPy, pandas, statsmodels
Retrieval, CV	FAISS, Qdrant, OpenCV, Tesseract
Tools	Slurm, Docker, AWS, FastAPI, Linux, Git
Languages	English (fluent), Vietnamese (native)